PIZZA HUT SALES ANALYSIS

INTRODUCTION

The **Pizza Hut Sales Analysis Report** is developed to provide a deep understanding of the company's sales dynamics across various channels, locations, and product categories. As one of the leading global pizza chains, Pizza Hut generates large volumes of sales data, which, when effectively analyzed, can reveal valuable insights into consumer behavior, market trends, operational performance, and revenue drivers.

This analysis encompasses:

- **Product Performance**: Assessing which pizza types (e.g., Veg, Non-Veg, Classic, Premium) and other menu items contribute the most to overall revenue.
- **Time-based Trends**: Identifying sales patterns based on time factors such as days, months, seasons, or holidays to help in demand forecasting.
- **Store-wise Analysis**: Comparing the performance of different branches or regions to highlight top-performing locations and identify underperforming areas.
- **Customer Insights**: Understanding preferences, repeat order trends, and pricing sensitivity to personalize offers and improve customer retention.
- **Marketing Impact**: Evaluating the effectiveness of promotions, combo offers, and campaigns on sales volume and average order value.
- **Operational Efficiency**: Highlighting factors that impact sales performance such as delivery time, order accuracy, and peak hours.

OBJECTIVE

The objective of this sales analysis is to:

- Understand overall sales performance across different time periods and regions
- Identify top-selling and low-performing products on the menu
- Analyze customer purchasing behavior and preferences
- Evaluate the effectiveness of promotional campaigns and offers
- Provide insights to improve marketing strategies and operational efficiency
- Support data-driven decision-making for revenue growth and customer satisfaction

PROBLEM STATEMENT

Pizza Hut has a diverse product range and a large volume of sales transactions. However, the company currently lacks a unified analysis that integrates data across product types, order volumes, and pricing to provide meaningful business insights. With no consolidated view of customer preferences, product performance, or order patterns, it becomes challenging to:

- Identify which pizza types and sizes are generating the most revenue
- Understand seasonal or daily order trends and peak sales periods
- Determine which categories (e.g., Chicken, Veggie, Classic) are most popular
- Evaluate the effectiveness of different pricing strategies across pizza sizes
- Optimize inventory and improve operational decisions based on demand

This analysis aims to bridge that gap by leveraging transactional and product-level data to uncover patterns, identify growth opportunities, and support data-driven strategic planning.

DATA SET DESCRIPTION

1. orders.csv
• Contains all customer orders with timestamps
• **Columns**: order_id, date, time
• Helps analyze order volume over time (daily, hourly, monthly)

2. order_details.csv
• Links each order to the pizzas purchased and their quantities
• **Columns**: order_details_id, order_id, pizza_id, quantity
• Enables revenue and frequency analysis per pizza

3. pizzas.csv
• Provides pricing and size information for each pizza
• **Columns**: pizza_id, pizza_type_id, size, price
• Useful for revenue calculations and understanding pricing strategy

4. pizza_types.csv
• Describes the pizza types, categories, and ingredients
• **Columns**: pizza_type_id, name, category, ingredients
• Helps group pizzas by category (e.g., Chicken, Classic, Veggie)

TOOLS AND TECHNOLOGIES FOR SALES ANALYSIS

Effective sales analysis at Pizza Hut relies on a robust suite of integrated tools and technologies that streamline data collection, processing, and visualization. These systems work in concert to provide real-time insights, enabling agile decision-making and operational optimization.

POS (Point-of-Sale) Systems

These systems are foundational for capturing real-time sales data, recording every transaction, item sold, and customer purchase. They provide the raw data necessary for all subsequent analyses, ensuring accuracy and immediacy.

BI & Data Visualization Tools

Tools such as Tableau or Power BI are crucial for transforming raw sales data into actionable insights. They allow for the identification of sales trends, seasonal patterns, and the visualization of key performance indicators (KPIs) through dashboards and reports, making complex data accessible.

Business Intelligence (BI) & Insights

Beyond mere visualization, BI platforms provide advanced analytical capabilities, including predictive modeling and prescriptive analytics. These insights support data-driven decisions, helping management understand market dynamics and anticipate future demand.

Customer Relationship Management (CRM) Systems

CRM systems are vital for understanding the customer base. They track customer preferences, purchase history, loyalty program participation, and feedback, enabling personalized marketing campaigns and improved customer retention strategies.

Inventory Management Systems

Efficient inventory management is critical for profitability. These systems monitor stock levels of ingredients and finished products, minimize waste due to spoilage, and automate reorder processes, ensuring that popular items are always in stock and operational costs are controlled.

Pizza Hut Integrated Systems

The synergy between POS, CRM, and inventory systems is paramount. An integrated approach ensures seamless data flow across all operational aspects, leading to enhanced efficiency, reduced manual errors, and a holistic view of the business for optimal sales performance and customer experience.

BASIC SALES ANALYSIS

Understanding the fundamental metrics of total orders and revenue is the first step in any comprehensive sales analysis. These figures provide a high-level overview of the business's activity and financial performance over a given period.

Total Order Count

The total number of orders processed stands at 21,350. This metric indicates the volume of transactions and customer engagement. A high order count suggests strong market demand and effective operational capacity to handle numerous customer requests. It's a direct measure of customer traffic and the operational efficiency in processing individual sales.

```
select count(order_id) as "Total_Orders"
from orders;
```

	Total_Orders
▶	21350

Total Revenue Generated

The total revenue generated is $817,860.05. This figure represents the total monetary value from all sales transactions. Revenue is a critical financial indicator, reflecting the overall success of sales strategies and pricing models. Analyzing revenue in conjunction with order count helps determine the average order value, providing insights into customer spending habits and the profitability of individual transactions.

```
select sum(quantity*price) as "Total_Revenue"
from order_details as od join pizzas as p
on od.pizza_id = p.pizza_id;
```

	Total_Revenue
▶	817860.049999993

Insight: The significant total order count combined with a robust revenue figure points to a healthy sales volume. Further analysis should focus on breaking down this revenue by product category, geographic region, and time period to identify specific drivers of success and areas for improvement. Understanding the average revenue per order ($817,860.05 / 21,350 = ~$38.31) can help in strategizing upselling and cross-selling initiatives to increase per-customer value.

MOST EXPENSIVE PIZZA AND ITS IMPLICATIONS

Identifying the most expensive menu item provides insights into premium offerings and customer willingness to pay. This information can influence menu design, marketing strategies, and ingredient sourcing.

The Most Expensive Pizza

```sql
select *
from pizzas
order by price desc
limit 1;
```

	pizza_id	pizza_type_id	size	price
▶	the_greek_xxl	the_greek	XXL	35.95

The most expensive pizza identified is "the_greek_xxl", priced at $35.95. This highlights a premium offering within Pizza Hut's menu, catering to customers seeking larger sizes or specialty ingredients.

Insight: The presence of a high-priced item like 'the_greek_xxl' suggests an opportunity for premiumization. Marketing efforts could emphasize the unique ingredients, generous size, or gourmet quality to justify its price point. Analysis should also consider the sales volume of this pizza relative to lower-priced options. If its sales volume is low despite the high price, it might indicate a need to either adjust pricing, enhance its perceived value through marketing, or re-evaluate its inclusion on the menu. Conversely, if it sells well, it confirms a segment of customers willing to pay for premium products, which could inform future product development.

Furthermore, understanding the demographics of customers who purchase this pizza could reveal valuable market segments for targeted advertising. This also opens avenues for bundle deals that include this premium pizza, potentially boosting overall revenue.

MOST ORDERED PIZZA SIZE: L DOMINANCE

Analyzing pizza size popularity is crucial for inventory management, production planning, and optimizing promotional strategies. The data clearly indicates a strong preference for a particular size.

The Most Ordered Pizza Size is L

The large (L) size pizza is the most frequently ordered, accounting for a significant 18,526 orders, which represents 38.10% of total pizzas sold. This dominance underscores its importance in Pizza Hut's product offerings.

```
select p.size as pizza_size,
COUNT(*) as most_ordered_pizzasize
from order_details od join pizzas p
on od.pizza_id = p.pizza_id
group by p.size
order by most_ordered_pizzasize desc
limit 1;
```

pizza_size	most_ordered_pizzasize
L	18526

Insight: The overwhelming preference for large-sized pizzas suggests that customers are often ordering for groups or value the perceived better value that larger pizzas offer. This has several implications:

- **Inventory and Production:** Operations should prioritize the availability of ingredients and efficient production lines for large pizzas. Stock levels for 'L' size crusts and packaging should be consistently high to meet demand and prevent stockouts, which could lead to customer dissatisfaction.
- **Marketing and Promotions:** Promotional efforts should heavily feature large pizzas. Bundle deals, family-sized meal packages, and discounts specifically on 'L' size pizzas are likely to resonate well with the majority of customers. Advertising campaigns could emphasize the value and shareability of large pizzas.
- **Menu Design:** While 'L' is dominant, it's important to understand the demand for other sizes (e.g., medium, small, personal) to ensure a balanced menu that caters to diverse customer needs (e.g., individual meals, smaller groups). However, 'L' should remain the cornerstone of the menu strategy.
- **Pricing Strategy:** The pricing of 'L' size pizzas should be competitive and reflect its high demand, potentially allowing for slight premium pricing if market conditions permit, or using it as a loss leader to attract more orders overall.

TOP FIVE POPULAR PIZZAS BY ORDER QUANTITY

Identifying the most popular pizzas is essential for understanding customer preferences, optimizing menu offerings, and focusing marketing efforts. These top performers are key drivers of sales volume.

Key Takeaways from Top Popular Pizzas

While specific names aren't provided, understanding that there are five distinct top-performing pizzas is crucial. These are the pizzas that consistently generate the highest order quantities, signaling strong customer demand and preference.

```
select
pt.name as pizza_name,
sum(od.quantity) as total_quantity_ordered
from order_details od join pizza_types pt
on substring(od.pizza_id, 1, length(od.pizza_id) - 2) = pt.pizza_type_id
group by pt.name
order by total_quantity_ordered DESC
limit 5;
```

pizza_name	total_quantity_ordered
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

Insight: The top five popular pizzas are the cornerstone of Pizza Hut's sales. It is imperative to:

- **Ensure Consistent Quality and Availability:** These pizzas must always be prepared to the highest standards and readily available. Any disruption in their supply or quality could significantly impact overall sales and customer satisfaction.
- **Strategic Placement on Menu and Promotions:** These popular items should be prominently featured on menus, in online ordering platforms, and in all marketing materials. Running special promotions or combo deals centered around these pizzas can further boost their sales and attract new customers.
- **Ingredient Forecasting:** Accurate forecasting for the ingredients used in these top five pizzas is vital for inventory management, reducing waste, and ensuring smooth operations. Supply chain efficiency directly impacts the profitability of these high-volume items.
- **Analyze Underlying Reasons for Popularity:** Further qualitative analysis can explore why these specific pizzas are so popular. Is it their flavor profile, value for money, historical popularity, or successful marketing? Understanding the "why" can inform future product development and menu innovation, by replicating successful elements in new offerings.
- **Cross-Selling Opportunities:** Leverage the popularity of these pizzas to cross-sell complementary items like sides, drinks, or desserts. Suggesting pairings at the point of sale can increase average order value.

INTERMEDIATE ANALYSIS

PIZZA QUANTITY BY CATEGORY AND AVERAGE DAILY ORDERS

Moving beyond basic metrics, this section delves into how pizzas are distributed across different categories and the consistent daily demand, providing a deeper understanding of sales patterns.

Pizza Quantity by Category

Analyzing pizza quantity by category (e.g., Classic, Specialty, Veggie, Meat Lovers) reveals which types of pizzas resonate most with customers. For instance, if 'Classic' pizzas like Pepperoni or Cheese show the highest quantity, it indicates a strong preference for traditional flavors.

Insight: Understanding category performance is critical for menu management and inventory. Categories with high quantities should receive prioritized resource allocation, from ingredients to marketing focus. Underperforming categories might need re-evaluation, either through menu adjustments, promotional pushes, or eventual removal. This breakdown also helps identify opportunities for innovation within popular categories or strategic expansion in niche categories.

```
select
pt.category,
sum(od.quantity) as total_quantity_ordered
from order_details od join pizzas p
on od.pizza_id = p.pizza_id join pizza_types pt
on p.pizza_type_id = pt.pizza_type_id
group by pt.category
order by total_quantity_ordered desc;
```

category	total_quantity_ordered
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

Average Daily Pizza Orders

The average number of pizza orders per day indicates the consistent demand for Pizza Hut's products. This metric is vital for staffing, operational planning, and capacity management. A stable average suggests reliable daily business volume.

Insight: A consistent average daily order count allows for more accurate forecasting of daily ingredient needs and staffing levels, minimizing waste and optimizing labor costs. Fluctuations from this average can signal external factors (e.g., holidays, local events) or internal issues (e.g., promotions, service disruptions). Monitoring this average helps in setting daily sales targets and evaluating operational efficiency on a day-to-day basis.

```
select
    round(avg(daily_total), 2) as avg_daily_pizzas
from(
    select
        o.order_date,
        sum(od.quantity) as daily_total
    from orders o join order_details od
    on o.order_id = od.order_id
    group by o.order_date
 )  as daily_pizzas;
```

	avg_daily_pizzas
▶	138.47

ORDER TRENDS BY HOUR: IDENTIFYING PEAK AND OFF-PEAK PERIODS

Understanding hourly order trends is crucial for optimizing staffing, managing kitchen efficiency, and tailoring promotional offers to specific times of the day. This analysis highlights when customer demand is highest and lowest.

Hourly Order Analysis

Analyzing order trends by hour reveals the daily rhythm of Pizza Hut's business operations. Typically, peak hours occur during lunch (12 PM - 2 PM) and dinner (6 PM - 9 PM) times, aligning with traditional meal schedules. Off-peak hours would include early mornings, late nights, and mid-afternoons.

Insight: Hourly order trends provide actionable intelligence for operational adjustments:

- **Staffing Optimization**: During peak hours, allocate more staff to kitchen operations, order taking, and delivery to handle the surge in demand efficiently. During off-peak hours, reduce staff levels to control labor costs, or assign them to preparatory tasks, cleaning, or training.
- **Kitchen Efficiency:** Prepare for peak times by pre-prepping ingredients and organizing workstations to maximize throughput. This reduces wait times and improves customer satisfaction during busy periods.
- **Targeted Promotions:** Implement special offers or discounts during off-peak hours to stimulate demand. For example, a "mid-afternoon snack" deal or late-night specials can encourage orders during typically slower periods, thereby increasing overall revenue without overwhelming staff during busy times.
- **Delivery Management:** Understand the hourly demand for delivery services to optimize driver schedules and routes, ensuring timely deliveries during peak times and efficient resource allocation during slower periods.
- **Marketing Timing:** Digital marketing efforts, such as push notifications or social media ads, can be timed to coincide with peak ordering windows to maximize visibility and encourage impulse purchases.

```
select
hour(order_time) as order_hour,
count(order_id) as total_orders
from orders
group by order_hour
order by order_hour;
```

order_hour	total_orders
9	1
10	8
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336
18	2399
19	2009
20	1642
21	1198
22	663
23	28

TOP THREE PIZZA TYPES BY REVENUE

Identifying the top-performing pizza types by revenue provides a crucial financial perspective beyond mere popularity by volume. These are the pizzas that contribute most significantly to the bottom line.

Analysis of Revenue-Driving Pizzas

While specific names are not provided, focusing on the top three pizza types by revenue is essential. These are likely a mix of highly popular items and premium-priced options that collectively generate the largest portion of sales income.

Identifying High-Value Pizzas

The pizzas that lead in revenue are not necessarily just the most ordered; they could be higher-priced specialty pizzas that, while perhaps ordered less frequently, contribute significantly more per sale. This highlights the importance of analyzing both volume and price.

Strategic Pricing and Promotion

For these top revenue generators, maintaining optimal pricing is crucial. Any promotional strategies should be carefully considered to avoid devaluing these high-contributing items. Bundling them with higher-margin sides or drinks can further enhance their profitability.

Quality and Consistency

Given their impact on revenue, ensuring the consistent quality and availability of ingredients for these pizzas is paramount. A negative experience with a high-value item could lead to significant customer dissatisfaction and a direct hit to revenue.

Insight: The top three pizza types by revenue are foundational to Pizza Hut's financial success. Marketing efforts should explicitly highlight these pizzas, emphasizing their unique selling points and value. Operational efficiency in preparing these specific pizzas should be optimized to maximize output and minimize costs. Furthermore, analyzing the profitability margins of these top revenue drivers can inform ingredient sourcing decisions and potential menu enhancements, ensuring that these high-performing items continue to contribute robustly to the company's financial health.

```
select pt.pizza_type_id,
sum(od.quantity * pz.price) as price_of_pizza
from order_details as od join pizzas as pz
on od.pizza_id = pz.pizza_id join pizza_types as pt
on pt.pizza_type_id = pz.pizza_type_id join  orders as o
on o.order_id = od.order_id
group by pt.pizza_type_id
order by price_of_pizza DESC
limit 3;
```

pizza_type_id	price_of_pizza
thai_ckn	43434.25
bbq_ckn	42768
cali_ckn	41409.5

PIZZA DISTRIBUTION BY CATEGORY: STRATEGIC INSIGHTS

Understanding the distribution of pizzas across various categories offers a nuanced view of customer preferences and helps in strategic menu planning, marketing, and inventory management.

Categorical Breakdown of Pizza Sales

Pizza Hut's menu typically includes several categories such as Classic, Specialty, Veggie, Meat, and perhaps seasonal or promotional offerings. Analyzing how total pizza sales are distributed among these categories provides critical insights into what segments of the menu are most popular.

```sql
select
pt.category,
count(distinct od.order_id) as total_orders
from order_details od join pizzas p
on od.pizza_id = p.pizza_id join pizza_types pt
on p.pizza_type_id = pt.pizza_type_id
group by pt.category
order by total_orders desc;
```

category	total_orders
Classic	10859
Supreme	9085
Veggie	8941
Chicken	8536

Insight: The distribution of pizzas by category holds significant strategic implications:

- Demand Alignment: If 'Specialty' pizzas, for example, constitute a large portion of sales, it indicates a customer base willing to explore diverse and often higher-priced options. Conversely, a strong preference for 'Classic' pizzas suggests a demand for familiarity and value.
- Inventory Management: Accurate category distribution data enables precise forecasting for specific ingredients and packaging relevant to each category. This minimizes waste and ensures that popular categories are always well-stocked, improving efficiency and reducing costs.
- Menu Innovation and Rationalization: Categories with high sales volumes can be areas for further menu innovation, introducing new variations or complementary items. Categories with low sales might be candidates for removal or re-evaluation, especially if they carry high operational costs or require specialized ingredients.
- Marketing Focus: Marketing campaigns can be tailored to promote specific categories. For example, if 'Veggie' pizzas are growing in popularity, targeted promotions towards health-conscious consumers or during specific dietary seasons (e.g., Lent) could be highly effective. If 'Meat Lovers' is dominant, emphasizing protein-rich options would resonate.
- Pricing Strategy: The distribution can also inform pricing strategies. Categories with higher demand might support slight price while those with lower demand might benefit from promotional pricing to stimulate interest.

ADVANCED ANALYSIS

REVENUE CONTRIBUTION BY PIZZA TYPE

"Revenue Contribution by Pizza Type" refers to **how much income each type of pizza generates** for a business (like Pizza Hut or Domino's). It shows the **share of total sales** that each pizza variety like Pepperoni, Margherita, Veggie Supreme, BBQ Chicken, etc brings in over a specific time period.

Insights: The Thai Chicken Pizza, The Barbecue Chicken Pizza, and The California Chicken Pizza are the top three revenue generators, collectively accounting for a significant portion of total pizza sales. This indicates a strong customer preference for chicken-based and classic pizza options. The Sicilian Pizza and The Greek Pizza also show substantial revenue contributions.

```sql
select
    pt.name as pizza_name,
    round(sum(od.quantity * p.price), 2) as pizza_revenue,
    round(sum(od.quantity * p.price) * 100 / (
        select sum(od2.quantity * p2.price)
        from order_details od2 join pizzas p2
        on od2.pizza_id = p2.pizza_id ), 2) as revenue_percentage
from order_details od join pizzas p
on od.pizza_id = p.pizza_id join pizza_types pt
on p.pizza_type_id = pt.pizza_type_id
group by pt.name
order by revenue_percentage desc;
```

pizza_name	pizza_revenue	revenue_percentage
The Thai Chicken Pizza	43434.25	5.31
The Barbecue Chicken Pizza	42768	5.23
The California Chicken Pizza	41409.5	5.06
The Classic Deluxe Pizza	38180.5	4.67
The Spicy Italian Pizza	34831.25	4.26
The Southwest Chicken Pizza	34705.75	4.24
The Italian Supreme Pizza	33476.75	4.09
The Hawaiian Pizza	32273.25	3.95
The Four Cheese Pizza	32265.7	3.95
The Sicilian Pizza	30940.5	3.78
The Pepperoni Pizza	30161.75	3.69
The Greek Pizza	28454.1	3.48
The Mexicana Pizza	26780.75	3.27
The Five Cheese Pizza	26066.5	3.19
The Pepper Salami Pizza	25529	3.12
The Italian Capocollo Pizza	25094	3.07
The Vegetables + Vegetable...	24374.75	2.98
The Prosciutto and Arugula ...	24193.25	2.96
The Napolitana Pizza	24087	2.95

CUMULATIVE REVENUE OVER TIME

This section presents the daily and cumulative revenue generated over time, providing insights into sales trends and overall financial performance.

Insights: The cumulative revenue shows a steady upward trend, indicating consistent sales growth over the observed period. Daily revenue fluctuates, with some days showing higher sales (e.g., January 8th with $2838.35) and others lower (e.g., January 4th with $1755.45). This suggests potential peak and off-peak days that could be further analyzed for staffing and promotional strategies.

```sql
select
    o.order_date,
    round(sum(od.quantity * p.price), 2) as daily_revenue,
    round(sum(sum(od.quantity * p.price)) over(order by o.order_date), 2) as cumulative_revenue
from orders o join order_details od
on o.order_id = od.order_id join pizzas p
on od.pizza_id = p.pizza_id
group by o.order_date
order by o.order_date;
```

order_date	daily_revenue	cumulative_revenue
2015-01-01	2713.85	2713.85
2015-01-02	2731.9	5445.75
2015-01-03	2662.4	8108.15
2015-01-04	1755.45	9863.6
2015-01-05	2065.95	11929.55
2015-01-06	2428.95	14358.5
2015-01-07	2202.2	16560.7
2015-01-08	2838.35	19399.05
2015-01-09	2127.35	21526.4
2015-01-10	2463.95	23990.35
2015-01-11	1872.3	25862.65
2015-01-12	1919.05	27781.7
2015-01-13	2049.6	29831.3
2015-01-14	2527.4	32358.7
2015-01-15	1984.8	34343.5
2015-01-16	2594.15	36937.65
2015-01-17	2064.1	39001.75
2015-01-18	1976.85	40978.6
2015-01-19	2387.15	43365.75

TOP THREE TYPES BY REVENUE IN EACH CATEGORY

This analysis identifies the top three pizza types by revenue within each category, highlighting the most popular and profitable options across different classifications.

Insights:

Chicken Category: The Thai Chicken Pizza, The Barbecue Chicken Pizza, and The California Chicken Pizza are the top performers, indicating a strong demand for chicken-based options.

Classic Category: The Classic Deluxe Pizza, The Hawaiian Pizza, and The Pepperoni Pizza lead this category, showcasing the enduring popularity of traditional choices.

Supreme Category: The Spicy Italian Pizza, The Italian Supreme Pizza, and The Sicilian Pizza are the top revenue drivers, suggesting a preference for bold and rich flavors.

Veggie Category: The Four Cheese Pizza, The Mexicana Pizza, and The Five Cheese Pizza are the top veggie options, with cheese-heavy and spicy vegetarian choices being most popular.

```sql
with PizzaRevenue as (
  select
    pt.category,
    pt.pizza_type_id,
    pt.name,
    sum(od.quantity * p.price) as total_revenue
  from order_details as od join pizzas as p
    on od.pizza_id = p.pizza_id join pizza_types as pt
    on p.pizza_type_id = pt.pizza_type_id
  group by
    pt.category,
    pt.pizza_type_id,
    pt.name),
      RankedPizzaRevenue as (
  select
    category,
    pizza_type_id,
    name,
    total_revenue
  from RankedPizzaRevenue
  where rn <= 3
  order by category,
  total_revenue desc;
```

category	pizza_type_id	name	total_revenue
Chicken	thai_ckn	The Thai Chicken Pizza	43434.25
Chicken	bbq_ckn	The Barbecue Chicken Pizza	42768
Chicken	cali_ckn	The California Chicken Pizza	41409.5
Classic	classic_dlx	The Classic Deluxe Pizza	38180.5
Classic	hawaiian	The Hawaiian Pizza	32273.25
Classic	pepperoni	The Pepperoni Pizza	30161.75
Supreme	spicy_ital	The Spicy Italian Pizza	34831.25
Supreme	ital_supr	The Italian Supreme Pizza	33476.75
Supreme	sicilian	The Sicilian Pizza	30940.5
Veggie	four_cheese	The Four Cheese Pizza	32265.70000000065
Veggie	mexicana	The Mexicana Pizza	26780.75
Veggie	five_cheese	The Five Cheese Pizza	26066.5

RECOMMENDATION

- **Promote Top Revenue-Generating Pizzas**
  – Focus marketing on *Thai Chicken*, *BBQ Chicken*, and *California Chicken*, which are the top 3 revenue contributors.

- **Bundle High-Revenue Pizzas in Meal Deals**
  – Create combo offers including best-selling pizzas to increase average order value.

- **Expand Chicken-Based Pizza Options**
  – Since most top-sellers are chicken-based, consider developing new chicken variants or limited-time offers.

- **Analyze Low-Performing Pizzas for Menu Optimization**
  – Identify and consider removing underperforming pizzas to streamline operations and reduce inventory costs.

- **Run Targeted Promotions for Mid-Tier Pizzas**
  – Boost revenue by promoting pizzas like *Hawaiian* and *Four Cheese* that are already doing moderately well.

- **Use Data for Regional Marketing**
  – Leverage local preferences (e.g., spicy or chicken-heavy pizzas) to tailor regional promotions.

- **Upsell with Toppings Based on Popular Pizzas**
  – Offer custom add-ons inspired by ingredients in top-selling pizzas like jalapeños, chicken, and bell peppers.

CONCLUSION

The comprehensive sales analysis of Pizza Hut reveals significant patterns and actionable insights that can be used to optimize business performance and enhance customer satisfaction. By examining order transactions, pizza types, sizes, and pricing structures, we've uncovered critical areas of strength as well as opportunities for growth.

• **Top-performing Products**: Certain pizzas, particularly those in the *Classic* and *Chicken* categories, consistently generate higher sales, indicating strong customer preference. These items can be further promoted to maximize revenue.

• **Size and Price Impact**: Medium and large-sized pizzas contribute more to total revenue than small sizes, highlighting customer willingness to pay more for larger portions. This can inform future pricing and combo strategies.

• **Category Insights**: Category-wise analysis showed that *Chicken* and *Deluxe* pizzas are among the most popular, while some *Veggie* and *Supreme* variants are underperforming.

• **Temporal Trends**: Sales exhibit notable peaks during weekends and meal hours (especially lunch and dinner), and during specific months or holidays. This insight is vital for staffing, inventory planning, and promotional timing.

• **Customer Behavior**: Analysis of ordering patterns shows that customers frequently purchase multiple pizzas in a single order, especially during peak periods. This behavior supports the effectiveness of combo offers and bundled deals.